Lagged Determinants of U.S. Output Growth

Ege Güney Kıymaç
Index No. 447987
e.kiymac@student.uw.edu.pl

Sung Kwan Chiu
Index No. 444585
s.chiu@student.uw.edu.pl

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Prepared under the supervision of Dr. A.W. Kolasa.

Abstract

This paper presents an autoregressive model of Real GDP with exogenous regressors that are connected to output in the Keynesian framework. A relatively standard/conventional statistical test suite is augmented with newly discovered modifications to the Kolmogorov–Smirnov test to facilitate time-series data and the interdependence brought by it. Building on the original validation methods that are rigorously tested over the decades, the paper manages to present moderate evidence that the estimator specified is efficient, well conditioned, and interpretable.

Contents

1 Introduction

1.1 Overview

Real GDP is a prominent measure of output due to its simplicity, interpretability, and effectiveness in communicating the overall activity of an economy. Alongside academic settings, Real GDP has retained its relevance in politics as one of the go-to measures for public communications regarding economic growth and prosperity. As per its wide-spread adoption, the measure has accumulated a plethora of literature referencing it from a multitude of perspectives.

This paper explores the link between GDP growth and theoretically grounded interactions of GDP deficit, unemployment gap, and output gap, alongside other exogenous regressors. Ultimately, the paper aims to show that empirical observations of Real GDP growth in the U.S. economy can be modelled optimally with a **Gauss–Markov** compliant **Distributed Lag Model** (DLM) specification where the exogenous contributors are theoretical descriptors of output.

1.2 Hypothesis

In the history of economic equilibria, multiple models have proposed mathematically defined relationships between Real GDP and many indicators. Moreover, some interactions between said indicators have been formalized into economic models. This paper focuses on testing the predictive and explanatory power of two such indicators involving unemployment gap, output gap, and GDP deflator (P_t^{GDP}):

- Unemployment Gap and Output Gap (**Okun**-like): $O_t = (u_t - u_t^*) \times (Y_t - Y_t^*)$
- Unemployment Gap and GDP Deflator (**Phillips**-like): $\Pi_t = (u_t - u_t^*) \times \Delta \log(P_t^{GDP})$

Alongside other exogenous regressors, this paper hypothesizes that a model of the form:

$$\Delta \log(Y_t) = C + \sum_{i=1}^{n} \beta_i X_{[i,\, t-1]} + \varepsilon_t \tag{1}$$

where $X_{[i,\, t-1]}$ includes the above defined interactions has an OLS estimator no strong evidence against **BLUE**.

By Okun's Law, we expect the unemployment gap to be modelled by the output gap and a negative coefficient. Therefore, the interaction O_t should be a redundant variable in the presence of its components (main-effects). On the other hand, the Π_t interaction is the only variable that brings an inflationary aspect into the model equation. Given the added information, we expect Π_t to add explanatory power though inflation while being supported by the in-set main effect of the unemployment gap. With these theorized expectations, build upon the design hypothesis and further hypothesize that O_t should have an insignificant, negative (if non-zero) coefficient while Π_t should have a significant and negative coefficient.

1.3 Theoretical Background

We consider two equations to base our selection of interactions. First, the **Phillips Curve** describes the relationship between inflation and unemployment. It can be expressed in terms of expected inflation and the unemployment gap as follows:

$$\pi_t = \mathbb{E}_t[\pi_{t+1}] - \alpha(u_t - u_t^*) + \varepsilon_t \tag{2}$$

where π_t is the inflation rate at time t, $\mathbb{E}_t[\pi_{t+1}]$ is the expected inflation rate at time $t+1$, $\bar{u}_t$ is the unemployment gap, α is a positive constant, and ε_t is a random error term. Equation 2 displays a negative relationship between inflation and unemployment gap with a factor of α. The inflation rate π_t in logarithmic space can be approximated by $\log(1+\pi_t) = \Delta\log(P_t)$ for sufficiently small π_t. (up to 8-10%) The presented interaction substitutes the deflator for price levels and uses $\Delta\log(P_t^{GDP})$ as a deflator-based inflation term. In principle, we expect (and hypothesize) that the pseudo-inflation term defined will again have a negative relationship with unemployment gap with a non-zero factor.

We can use the Phillips Curve to connect inflation and unemployment; however, the connection of this interaction to our regression target is not yet established. Utilizing the **Fischer Equation**, we can relate inflation to nominal interest rates, and express output as a function of inflation using the **IS Curve**.

$$\begin{aligned} r_t &= i_t - \pi_t \\ \Rightarrow Y_t &= Y^* - \alpha(r_t - r_t^*) \end{aligned} \tag{3}$$

Having established Y_t as a function of π_t, we can expect to see a relationship in growth rates as well.

Next, we consider **Okun's Law**, which describes the relationship between unemployment and output (Real GDP):

$$\frac{(Y_t - Y_t^*)}{Y_t^*} = -\gamma(u_t - u_t^*) \tag{4}$$

where Y_t is the actual output, Y_t^* is the potential output, γ is a positive constant, and $(u_t - u_t^*)$ is the unemployment gap. Okun's Law can be algebraically rearranged to express Y_t in levels:

$$\begin{aligned} \frac{Y_t}{Y_t^*} - 1 &= -\gamma(u_t - u_t^*) \\ Y_t &= Y_t^*(1 - \gamma(u_t - u_t^*)) \end{aligned} \tag{5}$$

The above transformation connects output to unemployment and establishes a negative relationship. A simple yet fundamental observation is that non-zero relationships of P_t, and $(u_t - u_t^*)$ are clearly defined in theory. However, we also observe that given direct relationship between unemployment gap and output gap is linearly defined without the need of the specific interaction O_t. In terms of gap modelling, we could conclude that these observations makes O_t directly redundant. Yet, in a growth rate context, we believe there's benefit in testing how the Okun based dynamics are mapped outside the gap framework. Interactions defined with these variables are therefore potential candidates to explain the output growth. (Although we expect one to be of no significance.) In any case, an empirical relationship can be analyzed to determine the differences between theoretical and observed interactions.

1.4 Literature Review

The literature review in this paper is aimed to justify two main choices made for our methodology and selection of variables:

- Selection of the exogenous variable set
- Usage of an ARX model

The theoretical relationships between exogenous variables and output were displayed in Section 1.3. In addition to said demonstration, there are numerous papers investigating these relationships. Primarily, "The Relationship between GDP and Unemployment Rate in the U.S." (Mandel and Liebens, 2019) employs a very similarly structured model to conduct correlation analysis. The paper demonstrates the existence of significant correlations $\rho_{Y,CPI}$ and $\rho_{Y,u}$ (among other variables) through multiple regression analysis. However, an important note regarding their application is the usage of CPI_{t-0} and u_{t-2} which do not align the orders selected in our paper.[1]

Another relevant paper, "GDP Forecasting: Machine Learning, Linear or Autoregression?" (Maccarrone et al., 2021), compares the performance of various simple regression models and machine learning algorithms in forecasting GDP. The authors conclude that ARX in their specification is the best model in "one-step-ahead" and comparable to the best in "multi-step-ahead" forecasts. This paper confirms the intuitively autocorrelated nature of GDP, and the fact that exogenous variables have the ability to improve fit quality.

Although there are many other papers investigating GDP modelling from simple regression to deep learning, the two papers mentioned above capture the reasoning behind the methodology design and model selection in our paper.

[1] Refer to Section **??** for the order selection of this paper.

2 Data

The initial model specification includes a wide range of macroeconomic indicators alongside the interactions. Observations for all series span from 1989Q4 to 2025Q1 and the frequency is quarterly. All series are sourced from the St. Louis Federal Reserve Economic Data (FRED) database.

Variable	Units	FRED ID
Real GDP (Y)	Billions of Chained 2017 Dollars (Adjusted)	GDPC1
Personal Consumption Expenditures (pce)	Billions of Chained 2017 Dollars (Adjusted)	PCECC96
Private Domestic Investment (pdi)	Billions of Chained 2017 Dollars (Adjusted)	GPDIC1
Government Expenditures ($govexp$)	Billions of Chained 2017 Dollars (Adjusted)	GCEC1
Output Deflator ($odef$)	Chained Index 2017=100 (Adjusted)	GDPDEF
Natural Unemployment Rate (u^*)	% (Not Adjusted)	NROU
Unemployment Rate (u)	% (Adjusted)	LRUN64TTUSQ156S
Potential Output (Y^*)	Billions of Chained 2017 Dollars (Adjusted)	GDPPOT

Table 1: Macroeconomic series used in the model along with their FRED IDs.

Using the series in Table 1, we construct our target variable and regressors as follows:

- Target Variable: $\Delta \log(Y_t) \times 100$
- Regressors:

$X_{[t,1]}$ (**pce**): $\Delta \log(pce_{t-1})$

$X_{[t,2]}$ (**pdi**): $\Delta \log(pdi_{t-1})$

$X_{[t,3]}$ (**$govexp$**): $\Delta \log(govexp_{t-1})$

$X_{[t,4]}$ (**$ugap$**): $\Delta(u_{t-1} - u^*_{t-1})$

$X_{[t,5]}$ (**$ogap$**): $\log(Y_{t-1}) - \log(Y^*_{t-1})$

$X_{[t,6]}$ (**$okun$**): $(u_{t-1} - u^*_{t-1}) \times ogap$

$X_{[t,7]}$ (**$phillips$**): $(u_{t-1} - u^*_{t-1}) \times \Delta \log(odef_{t-1})$

All variables that are observed in levels are transformed into log differences while both the gap variables and interactions are not differenced. (except for *ugap* due to multicollinearity concerns.) In our specification, log differencing is used to clean the regressors of slow-moving trends and potential multicollinearity; which is mostly an artifact of data in pure levels. The gaps and interactions are already detrended to an extent by their construction.

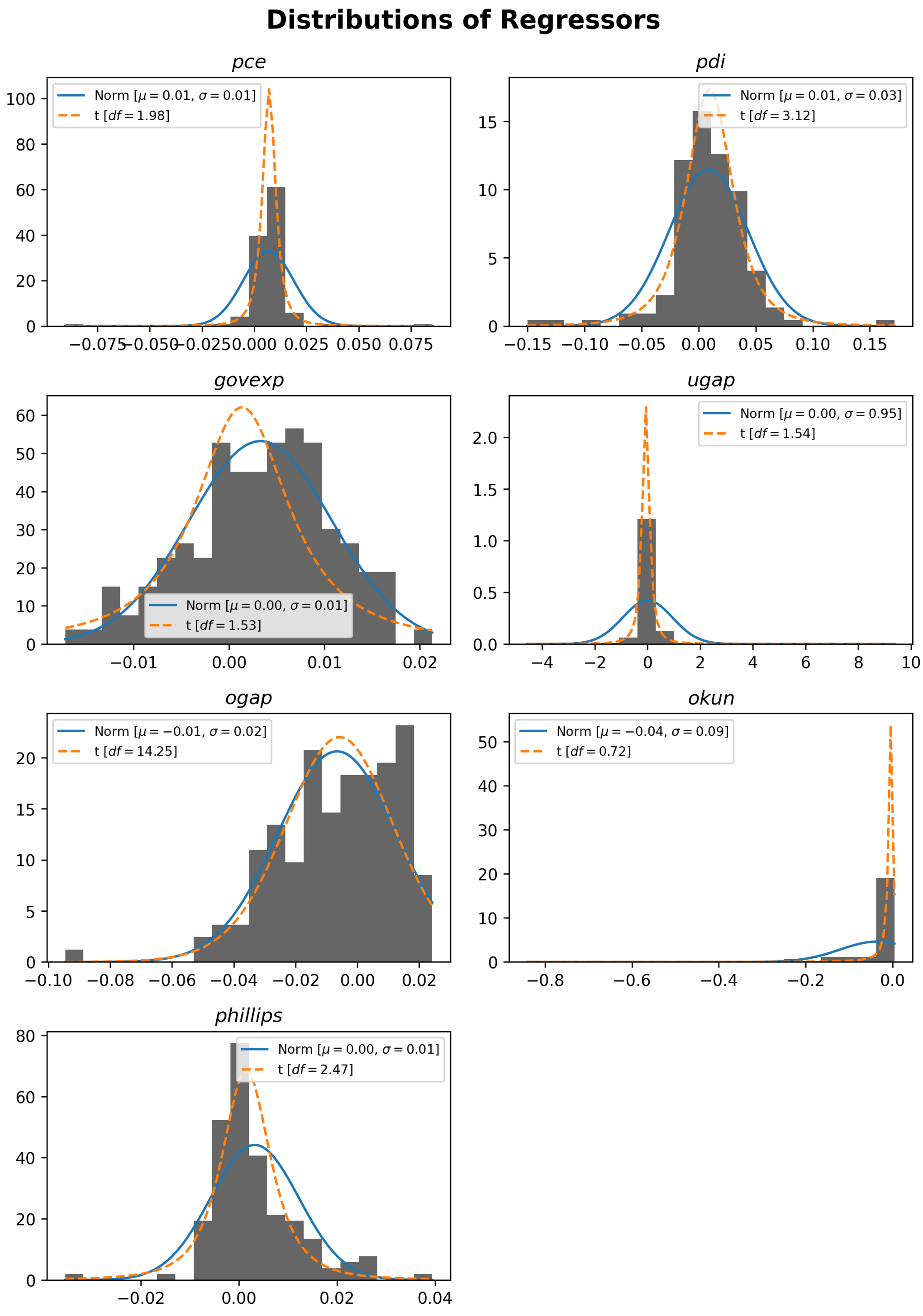

Figure 1: Density histograms of regressors.

3 Methodology

The proposed methodology of this paper can be split into four main components:

- Data Preparation and Examination
- Model Estimation
- BLUE Evaluation
- Post-Fit Validation

Each component is detailed in the following subsections.

3.1 Data Preparation and Examination

The preparation step consists of using the series defined in Table 1 to create regressors and target as listed. Vectorized transformations are applied to the raw data to create the necessary variables. The differencing process requires each observation to have a predecessor; therefore the first observation is lost in the transformation. This pushes our observation window from 1989Q4–2025Q1 to 1990Q1–2025Q1.

After the variables are constructed, we shift all regressors by one period to produce the final dataset. This ensures that all regressors at time t are known; making the model have predictive validity rather than being a purely explanatory specification. Having shifted the regressors, we again lose the first observation in the dataset, pulling the final observation window to 1990Q2–2025Q1 with a total of 140 observations.

3.1.1 Design Matrix Formation

X is formed by converting the model equation specified in Equation 1 into matrix form. The inner product operation of OLS estimation can be written as a matrix multiplication of observations and coefficients:

$$\underbrace{\begin{bmatrix} 1 & X_{1,2} & \cdots & X_{1,m+1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,2} & \cdots & X_{n,m+1} \end{bmatrix}}_{\text{Design Matrix}} \times \begin{bmatrix} C \\ \beta_1 \\ \vdots \\ \beta_{m+1} \end{bmatrix} \tag{6}$$

Where n is the number of observations and m is the number of features (including the intercept) Using our observation count $n = 140$ and number of regressors $m = 7$, the design matrix X takes the dimensions of 140×8.

3.1.2 Regressor Examination

While the model order will not be adjusted based on the results, the Cross Correlation Function (CCF) of each regressor with the target variable will be examined to confirm the appropriateness of the lag structure. The **Box–Jenkins Prewhitening** procedure will be applied to expose a clearer signal of the underlying cross-correlations. Box–Jenkins Prewhitening transforms the exogenous variables into residuals of their AR representation, then uses the β coefficient to re-scale the dependent variable. (Hackhard and Romer, 2025) Formally, the $AR(1)$ variant of the process can be shown as:

$$\begin{aligned} \hat{X}_i &= (L^1 X_i)\beta_i \\ \hat{y}_i &= y\beta_i \\ \rho_{y,i} &= \operatorname{corr}(\hat{y}_i, X_i - \hat{X}_i) \end{aligned} \tag{7}$$

Where L^1 is the lag-1 operator.

Additionally, the **Variance Inflation Factor** (VIF) of all regressors will be checked to ensure multicollinearity is not present. VIF is computed as:

$$VIF_{X_i} = \frac{1}{1 - R^2_{X_i}} \tag{8}$$

Where $R^2_{X_i}$ is the coefficient of determination of the regression of X_i on all other columns of X. The common threshold for multicollinearity, $VIF > 10$, will be used.

3.2 BLUE Evaluation

After estimating the model coefficients, we evaluate the Gauss–Markov conditions to confirm whether the resulting fit is BLUE. Previous stages of the methodology involve checks for the pre-fit Gauss–Markov conditions:

- **Linearity of Parameters:** by definition the model employed is a linear combination of regressors and coefficients. Therefore, this condition is satisfied at the stage of model selection.
- **Full Rank:** $\operatorname{rank}(X) = k$ must hold for $X^\top X$ to be invertible. Therefore, this condition must hold for the model to be estimable via OLS.

The remaining Gauss–Markov conditions concern ε and are verified post-estimation:

- **Spherical Errors:** This condition holds if $\operatorname{Var}(\varepsilon) = \sigma^2 I_n$; implies that the residuals are homoskedastic and uncorrelated. The two criteria can be tested separately using the **Breusch–Pagan Test** for homoskedasticity and the **Breusch–Godfrey Test** for autocorrelation.

- **Strict Exogeneity:** In textbook terms strict exogeneity requires $\mathbb{E}[\varepsilon_i \mid X] = 0$. However, this condition is not directly testable. It's often assumed to hold given weak exogeneity holds, and the design matrix X shows no direct signs of endogenous interactions. A **RESET Test** will be employed to check for model misspecification and omitted nonlinearities. Moreover, the **Harvey–Collier Test** will be used to further assess model specification. Strong exogeneity will then be assumed, and therefore any finding of BLUE will be referred to as "approximate" under the modelling assumptions.

3.3 Post-Fit Validation

After estimating the model and evaluating the Gauss–Markov conditions, regression coefficients and residuals will be evaluated. Alongside understanding variable significances and outliers, this setep will potentially highlight the sources of issues that may arise with the BLUE evaluation.

3.3.1 Coefficient Significance

Coefficients will be tested for significance both individually and jointly. Individual significance will be tested with ***t*-tests**, while model-wise joint significance will be tested with an ***F*-test**. Additionally, the joint significance of coefficient subsets will be tested using **Wald Tests**.

3.3.2 Residual Analysis

The residuals of a model are very important both in predictive terms, and as tools of analysing the model fit. Therefore, multiple aspects of the residuals will be examined before deeming the model valid/efficient.

To begin with, the fitted values will be plotted against the observed target to display the directionality and scale of residuals in a visual manner. This will be complemented by a QQ plot and a **Shapiro–Wilk Test** to assess the normality of residuals. Normality is not a Gauss–Markov condition, but it is important for inference.

Finally, potential outliers will be examined using **Leverage** and **Cook's Distance**. Given the nature of our target, we can immediately expect that the 2008 Financial Crisis and 2020 COVID-19 shocks will be visible in this portion of the analysis. If found to be detrimental to the model generalization, removal of outliers will be considered as required.

4 Results

4.1 Data Examination

With the design matrix constructed as per Section 3.1.1, we can compute the inter-regressor interactions with VIF and the regressor-target relations with CCF. Our initial VIF computations are as follows:

Regressor	**VIF**
pce	4.79
pdi	2.05
govexp	1.53
ugap	9.22
ogap	3.94
okun	4.62
phillips	3.22

Table 2: Variance Inflation Factor (VIF) of all regressors.

All VIF values are below the threshold, and we can conclude that multicollinearity is not a concern in our design matrix.

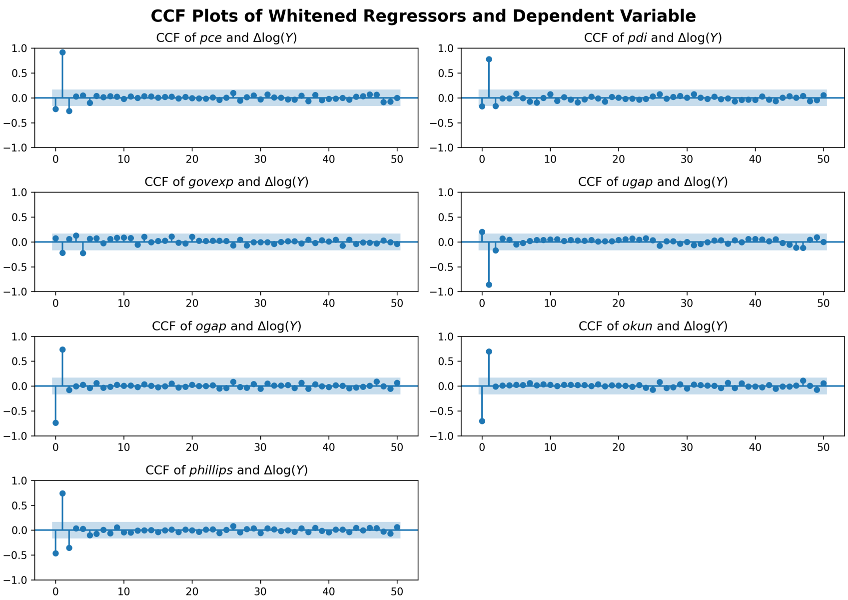

Figure 2: CCF plots of the whitened exogenous variables and the dependent variable.

The CCF results in Figure 4.1 show all regressors have significant lag-1 correlations, supporting the validity of the matrix specification with L^1. As expected, *pce* and *pdi* show strong positive correlations while *ugap* shows a negative correlation as theorized. Moreover, almost all variables with significant lags have a sign changed significant correlation immediately after the initial lag that's significant. This might be an indicator of cyclical behavior, or "response functions" with overshooting. *ogap* and *okun* show particularly similar magnitudes and sign patterns. With *ogap* being a component of *okun*, we can hypothesize that *okun*'s correlation pattern is mostly inherited from *ogap* rather than *ugap*. (at least in the first 2 lags)

4.2 Model Estimation

Having confirmed the design matrix is sufficiently well-behaved, we proceed with the estimation of the initial model. The fit results in the coefficients:

Variable	**Coefficient**	**t-test** p	**Significance**
C	0.0288	0.8309	X
pce	76.8300	$\approxeq 0$	✓
pdi	9.7348	0.0229	✓
govexp	-22.6115	0.0730	X
ugap	1.1538	$\approxeq 0$	✓
ogap	-5.0182	0.5201	X
okun	-3.4977	0.0670	X
phillips	-32.7522	0.0315	✓
			$\sum_{\checkmark} = 4$

Table 3: Estimated Coefficients of the Initial Model.

Table 3 shows a decent result with multiple coefficients being significant. We see that one of our interaction terms, *phillips*, has an individually significant coefficient while *okun* does not. The overall model fit statistics confirm this observation:

Statistic	**Value**
R^2	0.4071
Adjusted R^2	0.3703
RMSE	0.864
F-statistic	36.8205
p-value (F)	$\approxeq 0$
$X^\top \hat{\varepsilon}$	$-8.49 \times 10^{-14} \approxeq 0$

Table 4: Overall model fit statistics.

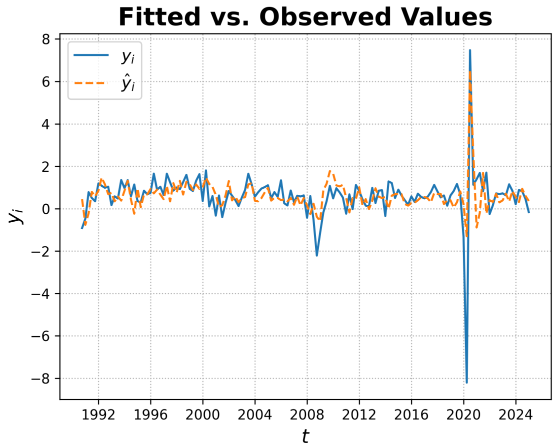

Figure 3: Fitted vs. Observed Values

With an R^2 of 40.71% and a passing F-test, the fit looks good at face value. However, the application of a RESET test tells a different story. The current model produces a p-value of 2.34×10^{-14} which is a strong indicator of misspecification. In fact, Figure 3 hints at a possible cause of this. With the 2008 Financial Crisis and 2020 COVID shocks in the data, we clearly see that the model goes out of its way to fit said points. This is especially visible in 2020Q3 where the model re-produces the extreme positive shock to output growth. This behavior is not expected from a generalized model and is potentially working against the model specification test. To confirm our suspicions of extreme outliers, we plot the Cook's Distance of all residuals:

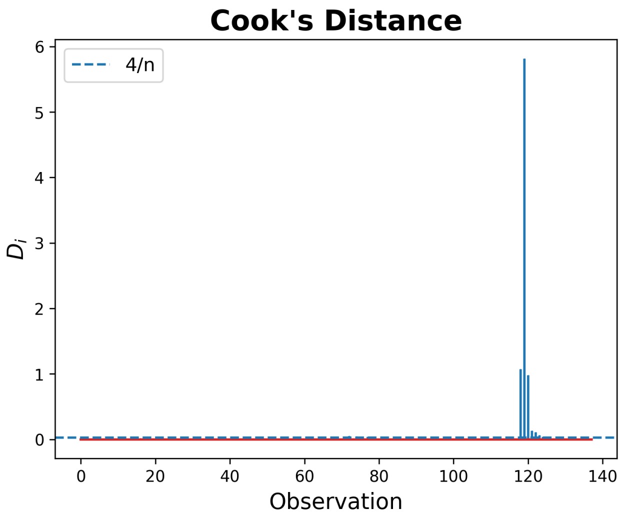

Figure 4: Cook's Distance of Initial Model Residuals.

We immediately see that the COVID-19 shocks dominate the influence statistics. $D_{2020Q3} = 5.81$ is orders of magnitude above the $4/n \approxeq 0.03$ threshold. In total, 10 observations exceed the threshold. 3 of these are attributable to the Financial Crisis (between 2008Q4 and 2010Q1) while the remaining 7 are consecutive observations from near-COVID (2020Q2–2021Q4).

In light of these results, it is clear that COVID-19 brings about undesirable fit-time behavior into the model. Of course, these observations are not to be discarded lightly, especially given time-continuity is crucial for preserving the interpretability of lagged models. However, given 7 consecutive observations with extreme influence on the model, there's a possibility of structural breaks. Therefore, two Chow tests were conducted with the break points at 2020Q2 and 2020Q3. Both tests confirmed the presence of structural breaks with $p_{2020Q2} = 0.01$ and $P_{2020Q3} \approxeq 0$. Seeing the results, we elected to re-estimate the current model using pre-COVID data before running further tests.

4.3 Pre-COVID Re-Estimation

After confirming the VIF of the pre-COVID design matrix cleared the thresholds ($VIF(X_i) < 10\, \forall X_i \in X$), the model was re-estimated and the following coefficients were obtained:

Variable	**Coefficient**	**t-test** p	**Significance**
C	0.1934	0.0528	X
pce	63.5334	$\approxeq 0$	✓
pdi	4.5738	0.0190	✓
$govexp$	-10.7911	0.1622	X
$ugap$	-0.1312	0.5661	X
$ogap$	-13.8038	0.0043	✓
$okun$	-0.1075	0.9322	X
$phillips$	-32.7953	0.0015	✓
			$\sum_{\checkmark} = 4$

Table 5: Estimated Coefficients of the Pre-COVID Model.

As in the previous estimation, Table 5 show four individually significant coefficients. Notably, *ogap* becomes significant while *ugap* loses its significance. The overall model fit statistics are as follows:

Statistic	Value
R^2	0.3955
Adjusted R^2	0.3507
RMSE	0.4525
F-statistic	8.83
p-value (F)	$\approxeq 0$
$X^\top \hat{\varepsilon}$	$-5.30 \times 10^{-14} \approxeq 0$

Table 6: Overall model fit statistics.

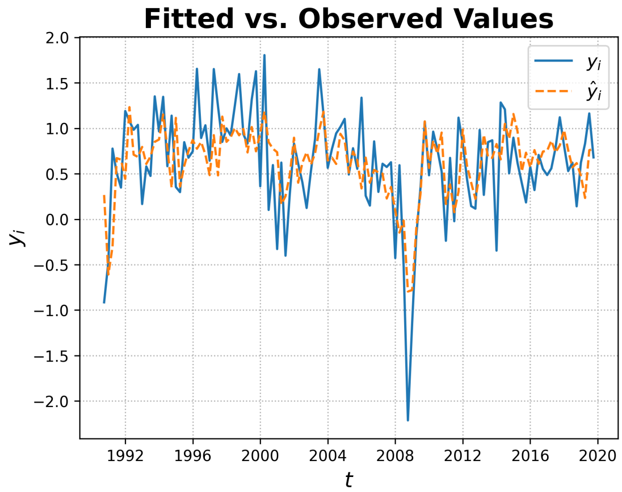

Figure 5: Fitted vs. Observed Values (Pre-COVID Model)

The fit statistics in Table 6 are comparable to the initial estimation, with a slight decrease in R^2. However, Figure 5 again displays the regime-breaking 2008 shock. Knowing the model specification issues of the previous fit, we confirm D_{2008Q4} is well above the $4/n$ threshold before running the RESET test again. The test again results in a rejection with $p = 0.0167$. Although this is again a misspecification indicator, we observe that a sizable improvement in p-value has been achieved without explicitly adjusting any regressors. However, the strategy of pulling back the observations in this case would lead to a large loss of data. After evaluating the 2008Q4 point as a break point candidate in a Chow test, we found $p = 0.1427$ and could not confirm a break. Therefore, we proceed with the pre-COVID model while introducing a signal dummy $I_{2008Q4} = \mathbb{1}_{\{t=2008Q4\}}$. By this approach, we're effectively treating 2008Q4 as a single-period shock. The dummy provides the model with a single coefficient to deal with the 2008 shock exactly while having no effect on other observations. On the other hand, this choice introduces a loss in degrees of freedom (DoF) by increasing the X dimension.

4.4 Shock Adjusted Model Re-Estimation

For one more time, we confirm the VIF of all regressors (including the newly added dummy) are below threshold and re-estimate the model to see if the loss of DoF is justified. The estimation results in the coefficients:

Variable	**Coefficient**	**t-test** p	**Significance**
C	0.2714	0.0059	✓
pce	54.3276	$\approxeq 0$	✓
pdi	4.8823	0.0088	✓
$govexp$	-9.5442	0.1942	X
$ugap$	0.0241	0.9135	X
$ogap$	-10.5652	0.0234	✓
$okun$	0.2778	0.8179	X
$phillips$	-24.2331	0.0157	✓
I_{2008Q4}	-1.7381	0.0006	✓
			$\sum_{\checkmark} = 6$

Table 7: Estimated Coefficients of the Pre-COVID Model.

Table 7 shows that the added dummy term is highly significant. Moreover, the addition has caused the intercept to become individually significant. The overall model fit statistics are as follows:

Statistic	**Value**
R^2	0.4577
Adjusted R^2	0.4121
RMSE	0.4286
F-statistic	10.0332
p-value (F)	$\approxeq 0$
$X^\top \hat{\varepsilon}$	$2.00 \times 10^{-14} \approxeq 0$

Table 8: Shock Adjusted model fit statistics.

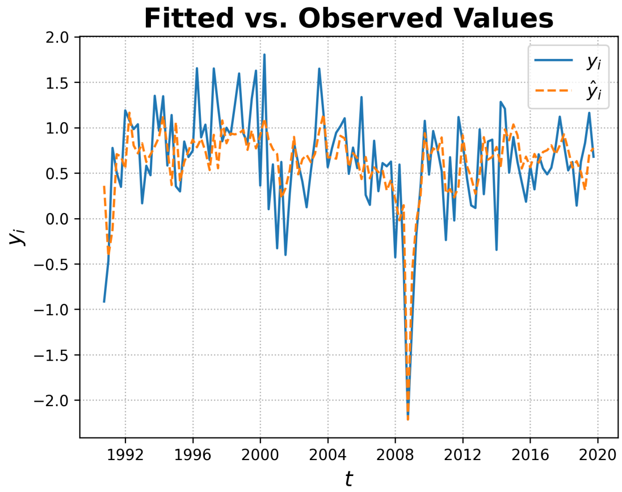

Figure 6: Fitted vs. Observed Values (Shock Adjusted Model)

The fit statistics in Table 8 show a sizable improvement in R^2 alongside an increase in the F-statistic. With the extravagant shock now accounted for, Figure 6 looks much better in terms of the presence of high-influence outliers. Plotting the Cook's Distance confirms this:

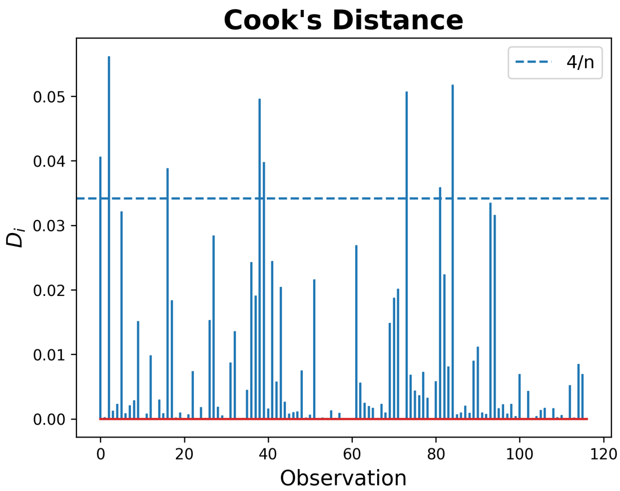

Figure 7: Cook's Distance of Shock Adjusted Model Residuals.

Although there still are 8 points above threshold, the distribution of D_i looks much more uniform and sensible. Running the RESET test one final time yields a p-value of 0.3837; indicating no evidence against correct model specification. With this confirmation, we proceed our analysis with the shock adjusted model.

4.5 Residual Diagnostics

Although the Cook's Distance plot displayed a basic look at the residuals, further diagnostics and tests are necessary to reach an accurate evaluation of the fit. Normality, Homoskedasticity, Autocorrelation, and Stationarity of residuals will be examined in this section.

4.5.1 Normality

Normality can both be tested for and examined visually. A QQ Plot alonside a Shapiro–Wilk test will be employed for this purpose.

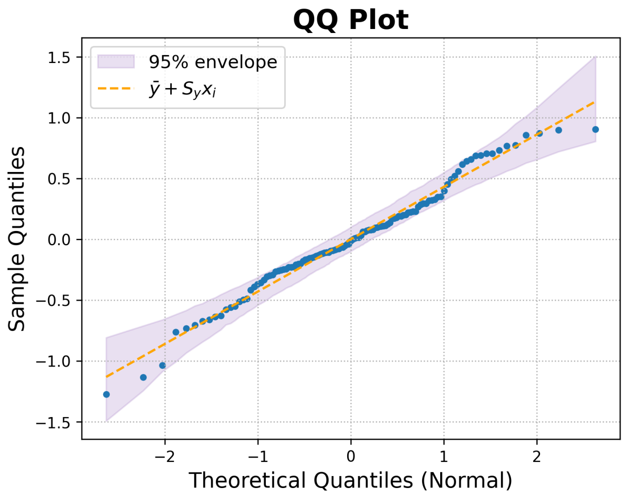

Figure 8: QQ Plot of Shock Adjusted Model Residuals.

The QQ Plot against normal quantiles shows that all residuals are contained within the 95% CI bands. This is a strong indicator of normality, which is further confirmed by the Shapiro–Wilk test results of $SW = 0.9825$ and $p = 0.1321$ failing to reject the null hypothesis of normality.

4.5.2 Homoskedasticity

Homoskedasticity is an important indicator of spherical errors and can be tested using the Breusch–Pagan test. A BP application on the shock adjusted model residuals yields $BP = 7.21$ and $p = 0.4853$; failing to reject the null hypothesis of homoskedasticity.

4.5.3 Autocorrelation

Autocorrelation of residuals indicates slow-moving patterns in the errors, which is undesirable and often results in inefficient estimates. To test for autocorrelation, we apply the Breusch–Godfrey test for lags $[1, 8]$ (2 years).

Lag	1	2	3	4	5	6	7	8
BG	5.00	5.57	6.93	8.57	8.67	9.84	10.19	10.77
p	0.975	0.938	0.926	0.927	0.877	0.868	0.822	0.785

Table 9: Breusch–Godfrey Test Results for Shock Adjusted Model Residuals.

All lags fail to reject the null hypothesis of no autocorrelation with strong p-values as seen in Table 9.

4.5.4 Stationarity

Finally, stationarity of residuals is an important aspect to consider in time series models. Non-stationary residuals can indicate omitted trends or cycles in the model specification. To test for stationarity, we apply the Augmented Dickey–Fuller (ADF) test on the residuals. The ADF test results in $ADF = -2.91$ and $p = 0.0036$; rejecting the null hypothesis of a unit root and indicating stationarity of residuals.

4.6 Gauss–Markov Conditions

Although we present strong evidence towards all Gauss-Markov conditions being satisfied, we further confirm our findings using the tests outlined in Section 3.2.

- **Linearity:** The model is linear in parameters by design (Equation 1).
- **Full Rank:** We computed a lack of perfect multicollinearity with $VIF(X_i) < 5 \; \forall X_i \in X$ and $\text{cond}(X) = 75.64$. While a $cond(X) < 30$ is the commonly used diagnostic threshold, having multiple interactions in X increases the expected condition number by design. Given the low VIF values, we accept the fulfillment of the full rank condition.
- **Spherical Errors:** We found no evidence against homoskedasticity, no autocorrelation up to lag 8, and stationarity of residuals. Therefore, we accept the fulfillment of the spherical errors condition.
- **Strict Exogeneity:** While strict exogeneity is not testable, the model specification tests and residual diagnostics were all in favor of correct model specification. While said tests are definitively not sufficient for the strong claim of exogeneity, we believe that the evidence presented is supporting weak exogeneity. Although further work would be necessary to confirm this, we will not deny the possibility of BLUE on the basis of this uncertainty.

5 Conclusion

With the help of normalization of variables, we were able to construct a well-behaved $ARX(1,1)$ model that, against expectations, showed no evidence against BLUE. Although the model showed signs of overfitting in-sample, the statistical tests and equality of variable distributions support the conclusion of an extremely interpretable estimator where coefficients may show connections to empirical elasticities of the used Keynesian regressors.

An unexpected finding was made with the unemployment coefficient, but the model as a whole shows enough stability and in-sample goodness-of-fit to warrant further comparison of empirical coefficients to economic theory. Moreover, we were able to display the use new and developing statistical tests and adapt to the unique challenges brought by the time-series context.

All in all, we believe that out model provides a decent starting point for further research and analysis. While the limited out-of-sample testing did not yield results that immediately display predictive instability, we strongly believe that as more data becomes available, the model will exhibit the well-known symptoms of overfitting. Therefore, we want to stress that the idea behind the creation of this model is to explore where theory and empirical evidence meet, and we encourage future researchers to build upon our findings.

6 Remarks

6.1 Remark 1: VIF Application on a Reduced Design Matrix

Although VIF just a measure and has no constraints on its application, the conventional use is to check for multicollinearity among the complete X matrix. In models with AR components, this application of VIF leads to two issues (assuming the dependent variable is actually autocorrelated):

- The AR components inside X will create processes equivalent to the an AR estimator of y. To demonstrate, let's assume a relationship $y_t \sim y_{t-1}$ exists in an $AR(2)$ model. As shifting the variable in time does not change the assumed relationship, we also have $y_{t-1} \sim y_{t-2}$. As a consequence, the design matrix (which is composed of y_{t-1} and y_{t-2}) will exactly replicate an $AR(1)$ estimator of y. With lags selected to maximize R_y^2, we end up also maximizing the R^2 in the VIF calculation.

- The point above extends to exogenous variables as well. Again, assuming $y_t \sim x_{t-1}$ exists, we also have $y_{t-1} \sim x_{t-2}$. Although $y_{t-1} \sim x_{t-2}$ never directly occurs, the design matrix in this case produces $y_{t-1} \sim x_{t-1}$. This is not an exact replication of the designed relationship, but by autocorrelation of y, it is a very safe assumption that $y_{t-1} \sim x_{t-1}$ will produce an R^2 comparable to the original estimator. As a result, we will again be maximizing the R^2 in the VIF calculation if we're constructing the model to explain the dependent variable as best as it can.

The points discussed show the well-known fact that VIF is not designed for these scenarios. Therefore, many studies, alongside this paper, elect to use VIF on a reduced X matrix stripped of AR components. (Niu and Li, 2022 & Karlström et al., 2023)

6.2 Remark 2: Time-Series Application of KS Test

The i.i.d. assumption of the classic KS test was addressed by Præstgaard with a bootstrapping approach. Yet, the examples, lemmas, and theorems presented in the paper were derived from row-exchangeable samples.

With row-exchangeability, Præstgaard was able to apply standard bootstrap resampling while preserving the continuity of the empirical distribution. However, time-series data inherently requires the row-structure to be preserved; hence rendering the standard bootstrap inappropriate. To preserve the time-dependent dynamics, our methodology elected to use a **Block Bootstrap** approach. This specific adaptation is not documented nor theoretically justified in Præstgaard's work; consequently, while we regard it as a reasonable and practically reliable extension for the purposes of this study, it remains an empirical choice that invites further theoretical investigation.

A block bootstrap 1-sample KS test was derived in a recent preprint paper (Chandy et al., 2025) and applications of bootstrap techniques in KS style tests were demonstrated in multiple studies. (Psaradakis, 2003 & Wyłupek, 2023) These works place bootstrap KS tests on a more solid theoretical footing, even though our specific use case is undocumented.

7 Bibliography

References in the Paper

Chandy, M., Schifano, E., Yan, J., & Zhang, X. (2025, November). Nonparametric Block Bootstrap Kolmogorov-Smirnov Goodness-of-Fit Test [arXiv:2511.05733 [stat]]. https://doi.org/10.48550/arXiv.2511.05733

Chen, X.-S., Kim, M. G., Lin, C.-H., & Na, H. J. (2025). Development of per Capita GDP Forecasting Model Using Deep Learning: Including Consumer Goods Index and Unemployment Rate [Publisher: Multidisciplinary Digital Publishing Institute]. *Sustainability*, *17*(3), 843. https://doi.org/10.3390/su17030843

Hackhard, B., & Romer, M. (2025, July). 9 Prewhitening; Intervention Analysis – STAT 510 — Applied Time Series Analysis. Retrieved November 28, 2025, from https://online.stat.psu.edu/stat510/Lesson09

Harvey, A. C., & Collier, P. (1977). Testing for functional misspecification in regression analysis. *Journal of Econometrics*, *6*(1), 103–119. https://doi.org/10.1016/0304-4076(77)90057-4

Karlström, A., Hill, J., & Johansson, L. (2023). Data-Driven Soft Sensors in Refining Processes – Pulp Property Estimation Using ARX - Models. *BioResources*, *18*(4), 8163–8186. Retrieved November 29, 2025, from https://ojs.bioresources.com/index.php/BRJ/article/view/22443

Maccarrone, G., Morelli, G., & Spadaccini, S. (2021). GDP Forecasting: Machine Learning, Linear or Autoregression? [Publisher: Frontiers]. *Frontiers in Artificial Intelligence*, *4*. https://doi.org/10.3389/frai.2021.757864

Mandel, D. C., & Liebens, P. (2019). The Relationship between GDP and Unemployment Rate in the U.S. *International Journal of Business and Social Science*, *10*(4). https://doi.org/10.30845/ijbss.v10n4p3

Niu, M., & Li, G. (2022). The Impact of Climate Change Risks on Residential Consumption in China: Evidence from ARMAX Modeling and Granger Causality Analysis [Publisher: Multidisciplinary Digital Publishing Institute]. *International Journal of Environmental Research and Public Health*, *19*(19), 12088. https://doi.org/10.3390/ijerph191912088

Præstgaard, J. (1995). Permutation and Bootstrap Kolmogorov-Smirnov Tests for the Equality of Two Distributions. *Scandinavian Journal of Statistics*, *22*(3), 305–322. Retrieved November 28, 2025, from https://www.jstor.org/stable/4616362

Psaradakis, Z. (2003). A Bootstrap Test for Symmetry of Dependent Data Based on a Kolmogorov–Smirnov Type Statistic [Publisher: Taylor & Francis _eprint: https://doi.org/10.1081/SAC-120013116]. *Communications in Statistics - Simulation and Computation*, *32*(1), 113–126. https://doi.org/10.1081/SAC-120013116

Riedl, A., & Wörz, J. (2018). A simple approach to nowcasting GDP growth in CESEE economies [Publisher: Oesterreichische Nationalbank (Austrian Central Bank)]. *Focus on European*

Economic Integration, (Q4/18), 56–74. Retrieved December 6, 2025, from https://ideas.repec.org//a/onb/oenbfi/y2018iq4-18b1.html

Rouksar-Dussoyea, B., Ming-Kang, H., Rajeswari, R., & Yin-Fah, B. C. (2017). Economic Crisis in Europe: Panel Analysis of Inflation, Unemployment and Gross Domestic Product Growth Rates. *International Journal of Economics and Finance*, *9*(10), 145. https://doi.org/10.5539/ijef.v9n10p145

Tang, L. (2020). Application of Nonlinear Autoregressive with Exogenous Input (NARX) neural network in macroeconomic forecasting, national goal setting and global competitiveness assessment [Number: 2005.08735 Publisher: arXiv.org]. *Papers*. Retrieved December 8, 2025, from https://ideas.repec.org//p/arx/papers/2005.08735.html

Wyłupek, G. (2023). A nonparametric test for paired data [Publisher: Elsevier]. *Journal of Multivariate Analysis*, *198*(C). https://doi.org/10.1016/j.jmva.2023.105229

References in the Source Code

Breusch, T. S. (1978). Testing for Autocorrelation in Dynamic Linear Models* [_eprint: https://onlinelibrary.wiley.com/doi 8454.1978.tb00635.x]. *Australian Economic Papers*, *17*(31), 334–355. https://doi.org/10.1111/j.1467-8454.1978.tb00635.x

David, D., & Wayne, F. (1979). Distribution of the Estimators for Autoregressive Time Series With a Unit Root. *Journal of the American Statistical Association*, *74*(366), 427–431. https://doi.org/10.2307/2286348

Durbin, J., & Watson, G. S. (1950). Testing for Serial Correlation in Least Squares Regression: I. *Biometrica*, *37*(3/4), 409–428. https://doi.org/10.2307/2332391

Godfrey, L. G. (1978). Testing for Higher Order Serial Correlation in Regression Equations when the Regressors Include Lagged Dependent Variables. *Econometrica*, *46*(6), 1303–1310. https://doi.org/10.2307/1913830

MacKinnon, J. (1996). Numerical Distribution Functions for Unit Root and Cointegration Tests. *Journal of Applied Econometrics*, *11*(6), 601–618. Retrieved November 21, 2025, from https://www.jstor.org/stable/2285154

Præstgaard, J. (1995). Permutation and Bootstrap Kolmogorov-Smirnov Tests for the Equality of Two Distributions. *Scandinavian Journal of Statistics*, *22*(3), 305–322. Retrieved November 28, 2025, from https://www.jstor.org/stable/4616362

Royston, J. P. (1982). An Extension of Shapiro and Wilk's W Test for Normality to Large Samples. *Journal of the Royal Statistical Society*, *31*(2), 115–124. https://doi.org/10.2307/2347973

Royston, J. P. (1995). Remark AS R94: A Remark on Algorithm AS 181: The W-test for Normality. *Journal of the Royal Statistical Society*, *44*(4), 547–551. https://doi.org/10.2307/2986146

Shapiro, S. S., & Wilk, M. B. (1965). An Analysis of Variance Test for Normality (Complete Samples). *Biometrica*, *52*(3/4), 591–611. https://doi.org/10.2307/2333709